# HESSAM BARASM MAMAGANI

**Senior Software Engineer | Full-Stack & AI Solutions Architect | Cloud & Enterprise Platforms**

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Location: Istanbul, Türkiye | LinkedIn | Portfolio | GitHub

Open to Sponsorship Opportunities in US/Canada/Europe | Willing to Relocate

## SUCCESS HIGHLIGHTS

- Sole architect & developer of 5 enterprise platforms (PilzPortal, WebRunner, InsightfulAI, EvenTracer, n8n AI) at Pilz Türkiye.
- Architected 1,190+ RESTful API endpoints across 102 controllers and 215+ EF Core tables with 99.9% uptime on Azure.
- Built full AI orchestration and RAG pipelines (OpenAI GPT + Anthropic Claude) with autonomous tool-calling and NL-to-SQL agents.
- Automated SAP ERP, CRM, and GİB e-Invoice operations via custom RPA, eliminating manual entry and cutting processing time by 80%.
- Published Smart Commit Generator AI on VS Marketplace and delivered 640+ hours of coding instruction to 150+ engineers.

## PROFESSIONAL SUMMARY

Senior Software Engineer and Full-Stack & AI Solutions Architect with deep expertise in designing and delivering production-grade enterprise platforms. Sole developer of Pilz Türkiye's core software ecosystem: an enterprise CRM/ERP platform serving all departments (PilzPortal), a hybrid SAP GUI & Playwright RPA engine (WebRunner), a multi-provider AI & RAG platform (InsightfulAI), and a multi-tenant B2B exhibition & executive BI SaaS powering 11 industrial symposiums, 2,525+ visitors, and 9,045 corporate accounts (EvenTracer). Proficient in C#, .NET 10, React 19, Next.js 16, TypeScript, Clean Architecture, CQRS, and Microsoft Azure, backed by engineering management and 640+ hours of technical instruction.

## TOP SKILLS

- **Full-Stack Development:** .NET 10, ASP.NET Core, C#, React 19, Next.js 16, TypeScript, Vite, MUI, Tailwind CSS
- **AI & LLM Integration:** OpenAI GPT, Anthropic Claude, Agentic Tool Calling, RAG Pipelines, Vector Stores, NL-to-SQL Agents, n8n
- **Cloud & Infrastructure:** Azure App Service, Container Apps, Blob Storage, Key Vault, SQL Server, PostgreSQL, Redis, Docker, CI/CD
- **Architecture & Patterns:** Clean Architecture, CQRS (MediatR), RESTful APIs, Entity Framework Core, SignalR, Background Workers
- **Security & Auth:** Federated JWT SSO, OAuth, FIDO2/WebAuthn (Biometric Passkeys), QR Auth, 5-Tier Role-Based Access Control
- **Enterprise Integration:** SAP GUI Scripting (COM Interop), Playwright RPA, Sovos/ISIS e-Invoice, Arvento GPS/OPET/HGS APIs, Word/PDF Gen

## PROFESSIONAL EXPERIENCE

### 1. Project Development Engineer (Senior Full-Stack & AI Solutions Architect)

**Pilz Türkiye (Pilz GmbH & Co. KG) | Aug 2024 – Present | Istanbul, Türkiye**

- Sole developer of 5 production platforms adopted across all Pilz Türkiye departments and external exhibitions.
- Architected and deployed ASP.NET Core APIs (1,190+ endpoints, 102 controllers, 215+ DB tables) on Azure and SQL Server.
- Built multi-provider AI orchestration integrating OpenAI GPT and Anthropic Claude with autonomous tools and RAG vector stores.
- Engineered SAP GUI and Playwright CRM automation, plus Sovos Foriba & ISIS SOAP Turkish Government (GİB) e-Invoicing.
- Delivered real-time SignalR hubs, WebPush notifications, Progressive Web Apps (PWA), and automated Word/PDF/Excel generation.
- Designed Federated JWT + FIDO2/WebAuthn biometric authentication with 5-tier RBAC and Azure Key Vault secret management.
- Integrated fleet and external APIs: Arvento GPS, OPET Fuel, HGS & HERE Maps Tolls, Google Maps, Twilio SMS, SendGrid.

#### Key Projects (Pilz & Open-Source)

**Pilz Portal – Enterprise Service Management Platform (CRM + ERP)** (pilztr.online/portal)

- **Stack:** ASP.NET Core 10 (804 endpoints, 62 controllers, 120 DB tables), React 19 (Vite), MUI, SQL Server, SignalR, Azure.
- **Features:** Consulting, Training, Engineering & Product lifecycles, Agentic AI quote tools, OCR onboarding, Sovos e-Invoice & fleet cost sync.
- **Impact:** Digitized all service and commercial workflows across Pilz Türkiye, improving operational efficiency by 70%+.

**WebRunner – Enterprise SAP GUI, Playwright RPA & e-Invoice Engine**

- **Stack:** .NET 8 Worker (31 RPA commands), .NET 4.8 x86 COM Interop Bridge (SapHelper), Playwright, WPF Scheduler, SignalR.
- **Features:** Automated SAP Sales Orders (VA01), Billing (VF01), Customer Master (XD01), FX/Aging reports, CRM sync, and AI failure diagnostics.
- **Impact:** Eliminated manual ERP/CRM data entry and reduced order-to-invoice processing time by 80%.

**InsightfulAI – Multi-Provider AI Knowledge & Document Analysis Platform** (insightfulai.online)

- **Stack:** ASP.NET Core (150 endpoints, 20 controllers), React 19 (Vite), MUI, OpenAI API, Anthropic Claude API, Azure.
- **Features:** RAG pipeline with vector stores, Assistants API, streaming chat, Machinery Safety AI Wizard, document Q&A, WebAuthn & QR login.
- **Impact:** Reduced technical and customer inquiry response time by 80% with AI-driven automation.

**EvenTracer – Multi-Tenant Exhibition, Phygital Campaign & Executive BI SaaS** (eventracer.com/en)

- **Stack:** Next.js 16, React 19, TypeScript, Tailwind CSS; .NET 10 Clean Architecture (CQRS/MediatR, 238 endpoints, 73 tables, Redis).
- **Features:** Multi-event YoY comparative BI & 2D pivot matrix, 18-slide C-Suite PPTX/PDF deck generator, Phygital VIP direct-mail QR & ElevenLabs AI voice campaigns, drag-and-drop form/badge builders, USB QR turnstile check-in, and booth lead scoring.
- **Impact:** Powered 11 exhibitions across 2,525+ visitors, 1,300+ check-ins, 9,045 CRM accounts, and 3,108 VIP QR invitations (4.9x B2B scan rate).

**n8n Enterprise AI Workflows & Smart Commit Generator AI** (marketplace.visualstudio.com)

- **Enterprise AI:** Dockerized n8n + PostgreSQL on Azure Container Apps hosting an NL-to-SQL Database AI Agent & ISO/EN safety standards tracker.
- **Open-Source:** Published VS Code & Open VSX extension (4,200+ LOC, 6 AI providers, 14 Git/PR commands, conflict dry-runs & secret scanners).

### 2. Full-Stack Software Engineering Instructor

**Üçüncü Binyıl Akademi | Aug 2023 – Aug 2024 | Istanbul, Türkiye**

- Conducted 640+ hours of live coding sessions on full-stack development across 4 ten-month cohorts (150+ students).
- Trained students in C#, ASP.NET Core Web APIs, React.js, SQL Server, Clean Architecture, and cloud integration.
- Guided students in architecting and building a full-stack e-commerce platform (Anazon) as a capstone project.

### 3. Chairman & Project Manager | Bid Manager | Senior Supervisor

**Earlier Career (Summarized) | Sep 2008 – Jul 2023 | Urmia**

- Managed operations, cross-functional teams, bidding, and engineering IT systems across construction and consulting firms.

## EDUCATION

- Master of Engineering (M.Eng.) in Structural Engineering | Urmia University (2015 – 2018)
- Bachelor of Engineering (B.Eng.) in Civil Engineering | Islamic Azad University (2003 – 2008)

## CERTIFICATIONS

- Fing IT Professional (Fing — Credential ID: ITPRO-18449071)
- C# for .NET Developers (Coursera)
- MCSE (Microsoft Certified Solutions Expert) & CCNA (Cisco Certified Network Associate)
- IELTS General Training

## LANGUAGES

English (Full Professional Proficiency) | Turkish (Fluent / Bilingual) | Azerbaijani, Persian (Native)